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Key Points:

- The competition between the formation of reaction products and fracturing produces steady-state values of permeability and porosity
- Steady-state conditions are determined by the dimensionless parameters that control the rates of fluid flow, reaction, and fracture growth
- A discrete rock model with pore-scale multiphysics reveals that a steady state threshold exists which defines clogging/fracturing regimes

Supporting Information:

Supporting Information may be found in the online version of this article.

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Unraveling the Controls of Rock Permeability Change During Volatile-Consuming Reactions

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Abstract Permeability changes of rocks are critical in hydration and carbonation, as they significantly affect the subsurface chemical composition. This study uses a one-dimensional reactive-transport-deformation model to define the dimensionless factors controlling permeability in the presence of rock fracturing, mineralization, and fluid flow. Furthermore, a coupled pore network-discrete element solver is used to validate the proposed parameters by capturing permeability evolution trends observed in the laboratory. Our analyses reveal a competition between pore clogging due to reaction product formation and pore dilation due to volume expansion and fracturing, showing the emergence of steady state in systems characterized by different initial porosity. Notably, two distinct regimes are found, one dominated by clogging and permeability loss, and the other controlled by fracturing, pore expansion, and permeability increase. These findings help rationalize the often contrasting trends resulting from field and laboratory evidence and can be used to enhance the efficiency of subsurface carbon storage.

Plain Language Summary This study aims to understand how hydration or carbonation affects the permeability of rocks and, in turn, the progress of these reactions in the subsurface. In this work, the key factors controlling permeability were first defined. Then, a discrete element model with pore-scale multiphysics was used to replicate permeability changes observed in experiments conducted on synthetic reactive rocks. The results show that a critical value of porosity, and in turn permeability, exists at which the competing volume change effects of solid product formation and expansion-induced fracturing reach a steady state. We also found that, in addition to the rates of the chemical reactions and fluid flow, these steady states depend in a crucial way on mechanical properties that are often ignored in these assessments, such as rock elasticity and fracture strength. These findings provide valuable insights to improve the efficiency of subsurface carbon storage by tailoring the injection strategy to the hydro-chemo-mechanical characteristics of the targeted rock formations.

1. Introduction

Decarbonization is an essential element of the global efforts to mitigate ongoing ecosystem deterioration. The most effective approach for reducing carbon dioxide in the atmosphere is to convert it into carbonate minerals, such as calcite and magnesite. Subsurface carbon storage in mafic and ultramafic rocks such as peridotite (Kelemen & Matter, 2008) and basalt represents a promising avenue to implement such solutions at a large scale. In fact, most hydration and carbonation reactions involve an increase in solid volume, which may lead to potential cracking (Nisbet et al., 2024). If reaction-induced cracking occurs, it is expected to accelerate the carbonation rate by increasing the rock permeability. However, it remains unclear which parameters control whether the reaction is sustained or inhibited. To guide the implementation of subsurface carbon storage in these formations, it is thus necessary to delineate the regimes of permeability evolution caused by volume expansion reactions.

Numerous laboratory-scale hydration and carbonation experiments have been conducted to replicate the subsurface storage of carbon dioxide. The typical rocks and minerals selected, along with their mechanical properties and volume change ratios, are shown in Table S1 in Supporting Information S1. Previous research has led to contrasting results in terms of permeability change. Specifically, the experiments inducing serpentinization and carbonation of peridotite exclusively led to a permeability decrease (Farough et al., 2016; Peuble et al., 2018), while those involving the hydration of periclase led to porosity-dependent results, with an increase of permeability in low-porosity samples (Uno et al., 2022; Zheng et al., 2018). Zhu et al. (2016) conducted carbonation experiments on a sintered olivine aggregate and found that nonuniform volume expansion introduces cracking inside the sample, which may expedite carbonation. However, Clark et al. (2018) found that the carbonation of cemented volcanic ash promoted permeability decrease despite crack formation within the sample. These

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apparent inconsistencies have led to numerous studies aimed at identifying the key controls of rock permeability change. Zheng et al. (2018) found that confinement is a key factor in controlling reaction rate and fluid flow. This study observed that when the confining pressure is larger than 30 MPa, the reaction rate is low and pore clogging becomes predominant during the experiment. Uno et al. (2022) showed that the ratio between reaction and flow rate (i.e., the Damköhler number) can control the permeability change and fracture during hydration. This indicates that permeability changes may result from two competing effects: product formation and expansion-driven fracturing. However, further research is needed to study the interplay between these factors and ascertain whether their competition underpins distinct regimes of permeability change. To do so, an essential task is to delineate whether a steady state separating these competing processes exists. Unfortunately, this is hindered by the constraints of laboratory-scale carbonation experiments, which typically cannot easily reach steady state because of their low reaction rate.

As an alternative, numerical models developed to emulate volume expansion reactions are being increasingly used. These models can be classified into two main categories: continuum (Evans et al., 2018, 2020; Malvoisin et al., 2021; Rudge et al., 2010) and discrete (Okamoto & Shimizu, 2015; Shimizu & Okamoto, 2016; Zhang et al., 2019). Rudge et al. (2010) proposed a 1D continuum model to capture various regimes of reaction-induced cracking, which identified the reaction rate as the key control of crack growth. Evans et al. (2020) used phase-field modeling to find that reaction-driven cracking can significantly enhance the transformation of olivine to serpentine. Similarly, Malvoisin et al. (2021) formulated an approach to mimic reaction-induced swelling which showed that a higher reaction extent is achieved when the chemically-driven mass is accommodated in the form of matrix deformation rather than through pore-filling mechanisms. To complement these approaches, discrete modeling has provided further insights. For example, a spring network model proposed by Jamtveit et al. (2008) indicates that stress heterogeneity is a crucial factor for microfracture networks and consequent permeability increases. Shimizu and Okamoto (2016) found that the transition between fracturing and non-fracturing behavior is governed by the Damköhler number. In this context, Zhang et al. (2019) simulated porosity evolution during volume expansion reactions, showing that their discrete model qualitatively matches the measurements. Despite these encouraging results, few studies clearly isolated the key controlling parameters of permeability change in volatile-consuming reactions by explaining their role in achieving steady state conditions.

In this work, we first identify the control parameters of the reactive transport in a deformable, brittle rock matrix, and then scale the sample's mechanical properties using the size effect law. Subsequently, a discrete element model with pore-scale multiphysics is built to test the validity of the identified control parameters and explore the existence of steady state conditions. By examining these factors in conjunction, this study aims to elucidate the mechanisms driving rock permeability change due to volatile-consuming reactions.

2. Dimensionless Control Parameters for Volatile-Consuming Reactions

A simple reaction, $rW + A = sB$, is considered whereby a mobile phase W (e.g., water, CO₂, or their mixture) reacts with an immobile solid A (e.g., periclase) to form a solid product B (e.g., brucite), and r , s are the stoichiometric coefficients. For simplicity, one-dimensional (1D) reactive transport is exclusively considered (Chen et al., 2024; Uno et al., 2022). To define the dimensionless control parameters, a reference time $t^* = 1/k_c$ is used to scale the equations, with k_c being a reaction rate constant. It follows that the 1D mass balance of reactive species and products can be written in dimensionless form as follows:

$$\phi_p \frac{\partial \left(\frac{P_f}{K_f} \right)}{\partial \left(\frac{t}{t^*} \right)} = \frac{K_f K_\phi}{k_c \mu_f L_a^2} \frac{\partial^2 \left(\frac{P_f}{K_f} \right)}{\partial \left(\frac{x}{L_a} \right)^2} + (s\nu_B - \nu_A - r\nu_W) c \quad (1)$$

where ϕ_p is the pore volume fraction, K_f the bulk modulus of the fluid, P_f the fluid pressure, K_ϕ the permeability, μ_f the fluid viscosity, L_a is the sample length, ν_i the molar volume of phase i , and c the concentration of solid A, determined by $c\left(\frac{x}{L_a}, \frac{t}{t^*}\right) = c_0 e^{-\left(\frac{t}{t^*} - \frac{P_f}{K_f}\right)}$ if the reaction starts at $t = t_0$, c_0 is the initial concentration.

The relationship between permeability and porosity can be established for a given sample under the homogenization-based Kozeny-Carman framework as revisited by Costa (2006).

$$K_{\phi} = C_c \frac{\phi_p^m}{(1 - \phi_p)} \quad (2)$$

where C_c is a factor playing a similar role to the Kozeny coefficient and m is the Archie exponent.

Another crucial component is the fracturing that occurs during volume expansion reactions. To simplify the analysis of fluid injection into an unsaturated or dry volume and approximate it as a diffusive process in a homogeneous (and, in this case, nominally fluid-saturated) domain, it is assumed that the reaction at a given location begins as soon as the fluid pressure increases beyond a minimal threshold value. For a given reaction under a set confining pressure, the nominal applied stress can therefore be expressed considering the dimensionless coefficient in Equation 1 (Rudge et al., 2010):

$$\sigma_N = \beta E \left(1 - \frac{c}{c_0} \right) = \beta E f \left(\phi_p, \frac{K_f K_{\phi}}{k_c \mu_f L_a^2}, \frac{P_{f,0}}{K_f} \right) \quad (3)$$

where β is a volume expansion constant specific for the considered reactive process, E is the Young's modulus of the rock, $P_{f,0}$ is the inlet fluid pressure.

This stress measure can be compared against the nominal tensile strength of the considered rock. In accordance with linear elastic fracture mechanics, tensile failure is governed by the toughness for mode I fracture, K_{Ic} . The additional quantities determining failure are nominal applied stress σ_N , sample length L_a , sample width b , and crack length a . By using size effect law (Bazant et al., 2021), the ensemble of dimensionless control parameters for volume expansion reactions can be therefore summarized as follows

$$g \left(\frac{\beta E \sqrt{L_a}}{K_{Ic}}, \phi_p, \frac{K_f K_{\phi}}{k_c \mu_f L_a^2}, \frac{P_{f,0}}{K_f}, \frac{a}{L_a} \right) = 0 \quad (4)$$

Where the same fluid type, fluid pressure, and bulk modulus are chosen at a constant confining pressure, with consistent sample length and reaction rate. The details of this derivation and the initial and boundary conditions can be found in the Text S1 in Supporting Information S1. Therefore, if the three dimensionless factors $\beta E \sqrt{L_a} / K_{Ic}$, ϕ_p , and $K_f K_{\phi} / (k_c \mu_f L_a^2)$ of different samples are the same, the crack patterns characterized by a / L_a are also expected to coincide, thus indicating that samples exhibit similar behavior. Additionally, porosity and permeability can evolve during the reaction according to Uno et al. (2022) which means that their values can change in the space of the dimensionless factors. However, given the difficulty of controlling or precisely quantifying the mechanical properties of rock cores in multi-physical experiments, hereafter synthetic tests based on a coupled pore network-discrete element solver will be used to further explore the role of the above nondimensional control parameters on rock permeability change during reactive transport. Since dimensional analysis captures only the first-order spatiotemporal features of hydro-chemo-mechanical processes, a numerical model is here used to test the applicability of the proposed dimensionless parameters under scenarios accounting for system nonlinearities.

3. Numerical Model

In this work, a numerical model is established as an alternative to laboratory or in situ tests. The model relies on a Discrete Element Method (DEM) enhanced with pore-scale multiphysics. The numerical specimens consist of 3,958 circular particles with radii ranging from 0.1 to 0.15 mm. The virtual experiments are based on a setup similar to that outlined by Uno et al. (2022), shown in Figure 1a. The rock deformation and fracturing processes are replicated with a standard DEM model based on bonded particles (Potyondy & Cundall, 2004), while a flow network model is adopted to simulate reactive transport (Fatt, 1956). To couple the two models, the pore network is constructed by placing its nodes at the center of the domains enclosed by loops of contacting particles (see Figure 1a). These network nodes are updated in conjunction with particle motion and consequent rock volume change. Based on Newtonian fluid flow in narrow channels, the flux q between two network nodes can be expressed through the Hagen-Poiseuille equation

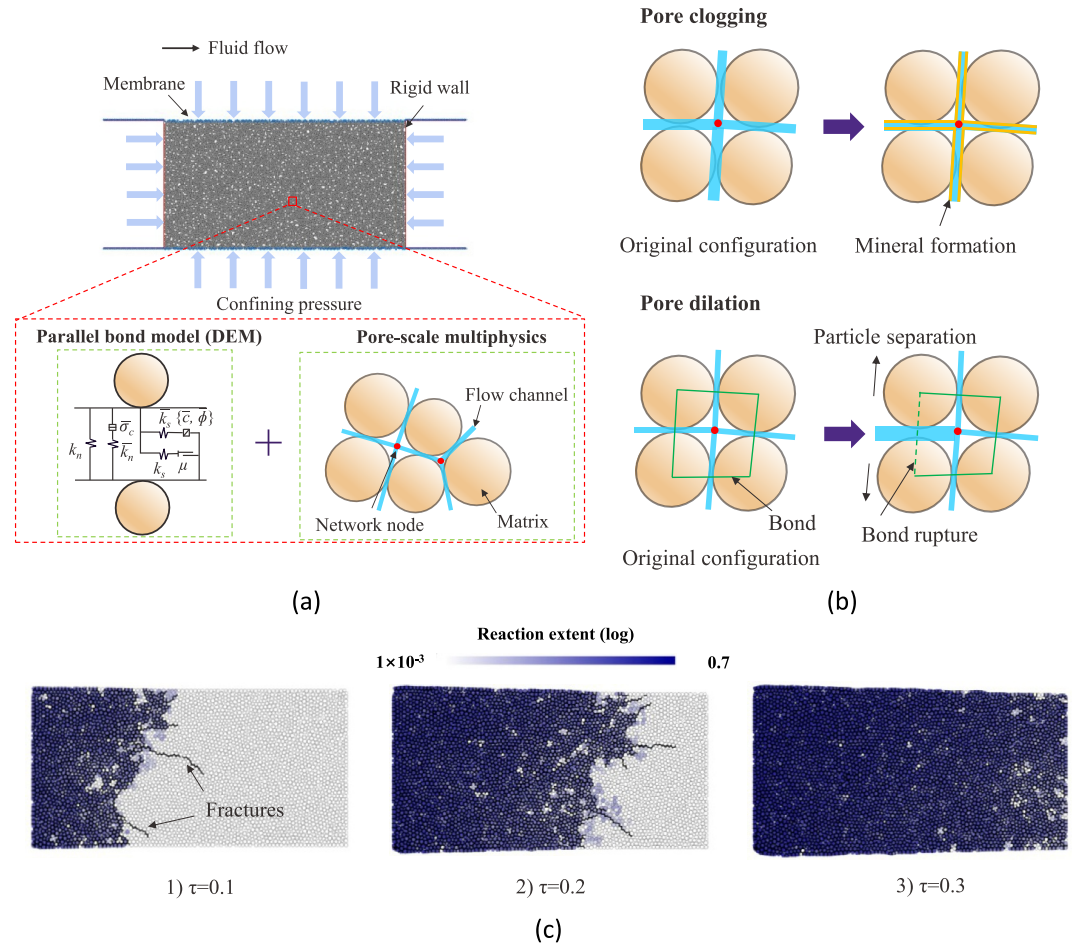


Figure 1. Schematics of (a) the numerical model used in this study, consisting of a Discrete Element Method model to simulate the mechanical behavior of the rock and a pore-scale multiphysics model to simulate reactive transport and (b) pore clogging and pore dilation models. (c) Examples of deformation, fracturing pattern, and reaction zone predicted by the numerical model for a dense specimen (initial porosity $\phi_p = 1.7\%$). The dimensionless time τ is defined as t/t^* , where $t^* = 1/k_c$.

$$q = \frac{w^3}{12\mu_f} \frac{\Delta P}{L_p} \quad (5)$$

where w is the width of the aperture (see Supporting Information S1 for its calculation), L_p is the length of the narrow channel, which is defined as the sum of the radii of two adjoining particles, ΔP is the pressure difference across the channel, and μ_f is the viscosity of the fluid.

In each domain, the change in fluid pressure ΔP is obtained from the continuity equation

$$\Delta P = \frac{K_f}{V_c} \left(\sum q \Delta t - \Delta V_{r_fluid} \right) \quad (6)$$

where $\sum q$ is the total inflow from the surrounding channels into one domain, Δt is the timestep, K_f is the bulk modulus of the fluid which controls the compressibility of the fluid, V_c is the volume of the domain, and ΔV_{r_fluid} is the volume change of the fluid participating in the reaction (see Supporting Information S1 for further details about its computation).

The porosity of the rock model is defined as

$$\phi_p = \frac{\sum V_c}{V_{\text{total}}} \quad (7)$$

where $\sum V_c$ is the total volume of the flow channel and V_{total} is the total volume of the simulated specimen.

Rigid boundaries are used for the left and right sides of the simulated specimen, while soft boundaries are adopted to mimic deformable membranes delimiting the other sides. A confining pressure is applied to the simulated membranes to impose a stress compatible with the target in situ conditions. In this study, the linear parallel bond model is adopted to simulate the tensile strength of rocks (Potyondy & Cundall, 2004), while a pore network model is used to simulate fluid flow (Fatt, 1956) and capture pore size alterations caused by expansion, clogging, and fracturing (Figure 1b). Both models are implemented in the software PFC (Itasca Consulting Group, Inc., 2021; Zhang et al., 2019). Further details about the model and its implementation can be found in Text S2 in Supporting Information S1. The model parameters are calibrated with reference to mechanical properties reported in the literature (e.g., Zhang et al., 2019) and summarized in Table S1 in Supporting Information S1. The main reactive process considered for the analysis is the hydration of periclase to brucite [$\text{MgO} + \text{H}_2\text{O} = \text{Mg}(\text{OH})_2$]. The complete set of parameters can be found in Table S2 in Supporting Information S1 and was verified by matching both the uniaxial compressive and tensile strength of periclase (Figure S2 in Supporting Information S1) and the porosity-permeability relation of unreacted rock specimens (Figure S3 in Supporting Information S1). For the latter, the results indicate a strong idealized correlation between the two variables, consistent with commonly used fitting equations (e.g., Costa, 2006). Further details about the calibrations of the hydraulic and mechanical model performance are provided in Text S3 in Supporting Information S1. The simulation results provide computed values of porosity, which are a function of the calibrated geomechanical properties (i.e., Young's modulus E , toughness K_{IC}) and prescribed chemical and hydraulic parameters (i.e., reaction rate k_c , dilation coefficient β , fluid bulk modulus K_f , fluid viscosity μ_f). These quantities are then plotted within the dimensionless spaces introduced by the previous section. Despite the presence of nonlinearity in the discrete model, the proposed dimensionless groups satisfactorily explain the convergence toward steady state. This finding suggests that nonlinear effects play a limited role in the systems examined in this study, thus rendering the proposed set of dimensionless parameters a viable option to interpret evidence from both field and laboratory measurements. The typical chemo-mechanical behavior emerging from the proposed model is displayed in Figure 1c, which shows selected examples of computed specimen deformation, fracture patterns, and reaction extent at specific reaction times. Further examples for different values of initial porosity are provided by Figure S4 and Text S4 in Supporting Information S1, showing that while the reaction front propagates more slowly in low-porosity samples due to their low initial permeability, the onset of fracturing can effectively augment the reaction rate.

4. Numerical Results and Comparison With Laboratory Evidence

This section illustrates the numerical results and compares them against those reported in laboratory tests. Figure 2a shows the porosity change of numerical samples S1-4 with a modulus close to periclase (302 GPa). The results indicate that if the initial porosity is higher than the critical value, the sample porosity decreases. Conversely, if the initial porosity is lower than the critical value, the sample porosity is predicted to increase. Additionally, two more sets of samples with intermediate (30 GPa) and low modulus (6 GPa) are also chosen for simulations in Figures 2b and 2c. The range of moduli from 6 to 302 GPa covers most types of natural rocks and minerals (see Table S1 in Supporting Information S1). Critical values of porosity can still be defined in Figures 2b and 2c. This indicates that while the presence of a threshold porosity is a consistent trait of these systems, its value is affected by mechanical properties such as the Young's modulus of the rock matrix. Given the relationship between porosity and permeability established in Figure S3 in Supporting Information S1, this suggests that the numerical model's conclusions align with experimental observations shown in Figure 2d. This hydration experiment by Uno et al. (2022) reveals a clear decrease in the permeability of the most porous periclase (MgO) sample, while the permeability of the densest sample increased by nearly three orders of magnitude. The permeability of the periclase sample with intermediate porosity displayed instead non-monotonic trends, with its value initially dropping by two orders of magnitude, and then increasing by up to two orders of magnitude above the initial value. The final permeability values of the six samples fall within a relatively narrow range between 10^{-17} and 10^{-16} m^2 . Based on these results, the permeability decrease indicates that pore clogging dominates the process, while the permeability increase reveals that fracturing can be the dominant process. In the numerical

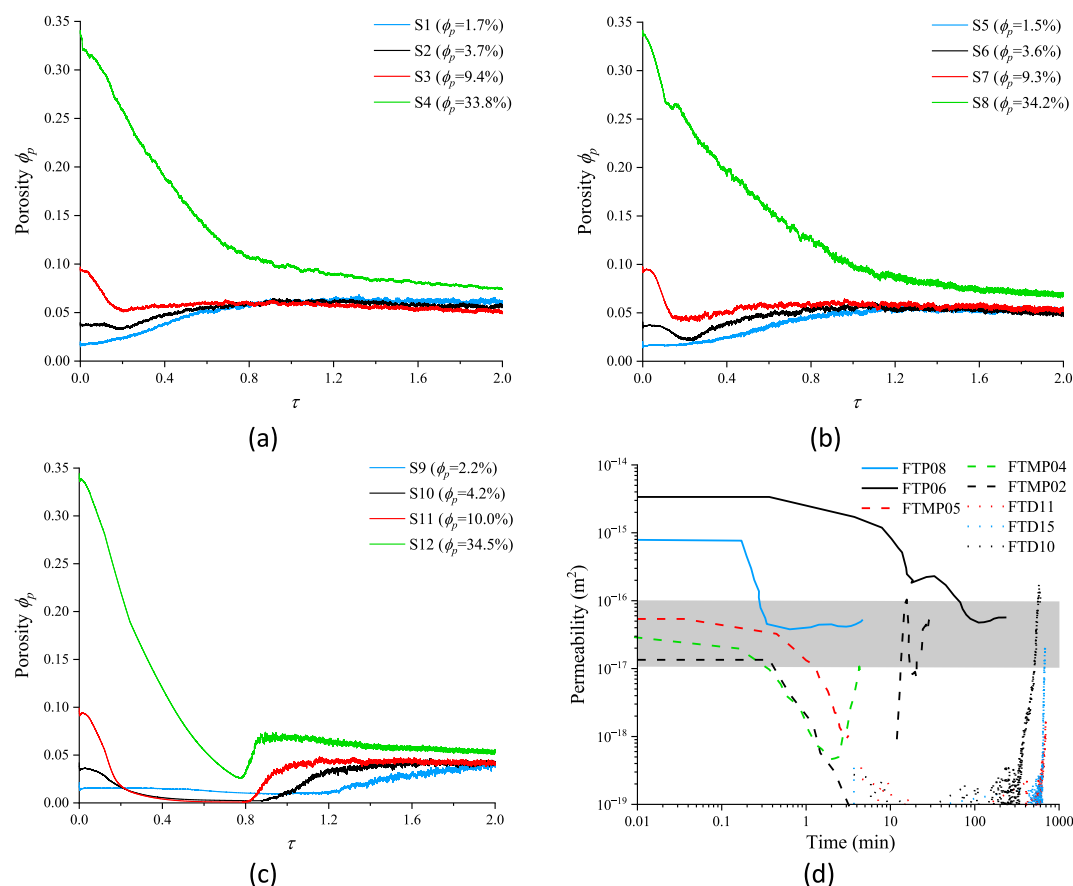


Figure 2. Porosity evolution trends for samples with different values of Young's modulus (a) close to periclase (302 GPa), (b) intermediate (30 GPa), and (c) low (6 GPa). (d) Evolution of the permeability of periclase (MgO) samples with different initial porosity, ϕ_p , reported by Uno et al. (2022). FTP denotes the high-porosity MgO sample, FTMP the intermediate-porosity MgO sample, and FTD the low-porosity MgO sample.

model, the reaction extent continues increasing once the porosity reaches the critical value, as shown in Figure S5 in Supporting Information S1. The numerical model effectively captures the phenomena of pore dilation and clogging considering aperture changes caused by reaction product formation and fracturing. Reaction product formation can decrease the permeability, while fracturing can increase it. The results outlined here also complement existing experimental studies on hydration and carbonation, which often overlook the crucial influence of the elastic modulus of the rock or focus exclusively on a specific rock type without comparative analysis with formations of differing stiffness.

To further examine the model results, the numerical sample is divided into 10 slices to examine the distributions of the reaction extent, porosity, and normalized pore pressure across the numerical samples of different initial porosity. A comparative analysis of samples S1-4 is presented in Figure 3, while additional comparisons are available in the Supporting Information S1 (Figures S6 and S7 in Supporting Information S1). The results indicate that the sample with the highest porosity (S4) exhibits a more uniform reaction extent at the onset of the synthetic test compared to the lowest-porosity sample. The distributions of the extent of reaction and normalized pore pressure become more homogeneous over time as the porosity and permeability evolve. The reaction of the high-porosity sample slows down, in that the pore clogging caused by reaction product formation is dominant, leading to a decrease in pore aperture and permeability. Conversely, S1, characterized by the lowest porosity, exhibits a significantly delayed reaction front at normalized time $\tau = 0.1$ compared to the other samples. However, as the reaction continues, fracturing becomes dominant, leading to porosity increase, larger interstitial aperture, and higher permeability. Most notably, a close inspection of the spatial profiles for all the samples clearly indicates very similar porosity profiles, which are also consistent with similar values of permeability.

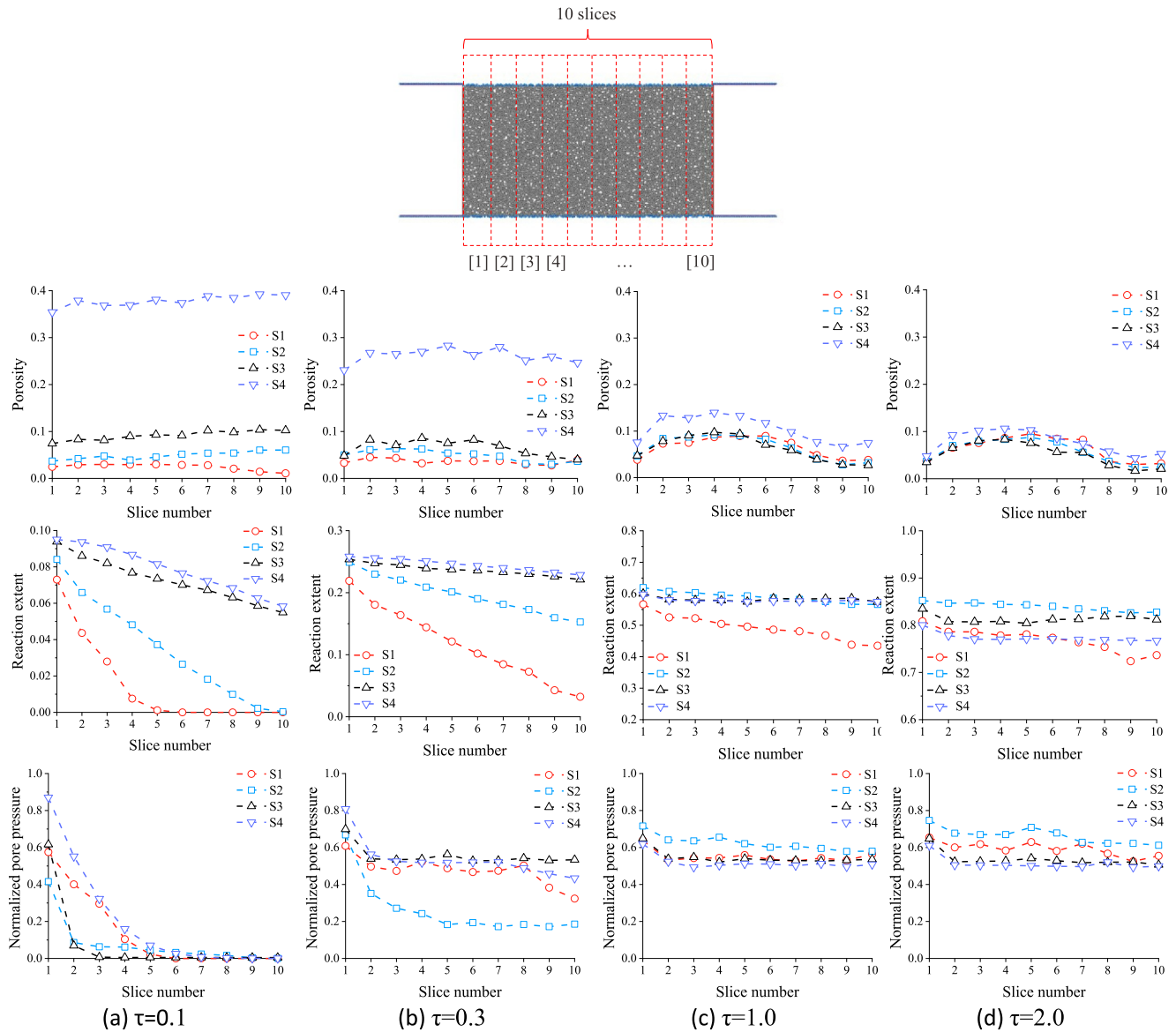


Figure 3. Comparison of the distributions of the porosity, reaction extent, and normalized pore pressure across numerical samples S1 (initial porosity $\phi_p = 1.7\%$), S2 (3.7%), S3 (9.4%), and S4 (33.8%) at dimensionless times $\tau = (a) 0.1, (b) 0.3, (c) 1.0, (d) 2.0$. The normalized pore pressure is defined as P_f/P_{f0} .

5. Discussion About Steady State

Based on the results shown in Section 4, newly formed reaction products can lead to pore clogging, while fracturing can facilitate pore dilation. The overall change in porosity or permeability results from the competition between these effects. This suggests that a steady state domain may exist toward which the permeability tends to converge. To explore this hypothesis, the initial and final states of all the samples are plotted in Figure 4a to capture the steady state locus in $\beta E \sqrt{L_a}/K_{lc} - \phi_p$ space. The results show a steady state threshold dividing the parameter space into two zones. This steady state was not explicitly identified in prior experimental works, in that data are often presented exclusively in terms of reaction rate and flow rate, with no effect of the mechanical properties of the rock. Our findings suggest instead that such locus exists and is characterized by a nonlinear trend that reflects the coupling between the system's mechanical (e.g., the Young's modulus and fracture strength) and chemo-hydraulic properties (e.g., permeability and reaction rate). Specifically, initial states located above the steady state locus are indicative of systems with relatively large permeability, in which expansion can be

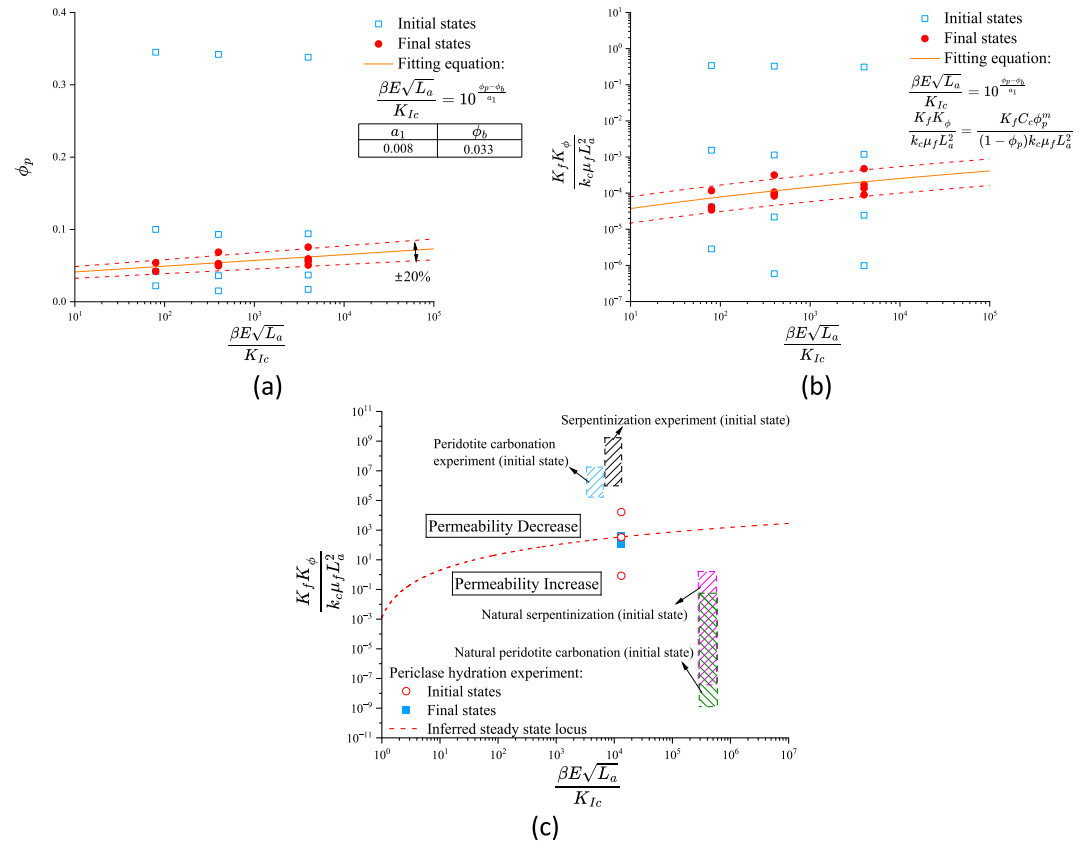


Figure 4. Steady-state loci of volatile-consuming reactions reported in (a) $\beta E \sqrt{L_a}/K_{Ic} - \phi_p$ space and (b) $\beta E \sqrt{L_a}/K_{Ic} - K_f K_\phi / (k_c \mu_f L_a^2)$ space. (c) Inferred steady state locus for typical hydration and carbonation experiments. Data sources: periclase hydration experiment (Uno et al., 2022); serpentinization experiment (Farough et al., 2016); peridotite carbonation experiment (Peuble et al., 2018); natural peridotite carbonation (Kelemen & Matter, 2008); natural serpentinization (Lamadrid et al., 2021; Uno et al., 2022). The sample length in natural peridotite carbonation and natural serpentinization is assumed to be 100 m which falls in the range proposed in Chen et al. (2024).

uniformly accommodated across the volume, ultimately reducing the potential for fracturing. In such cases, pore clogging tends to be dominant, leading to a decrease in permeability. Conversely, if the initial state is located below the steady state locus, the low flow rate implies that expansion can only occur at the front of the reaction process, then rendering reaction-driven cracking more significant due to its capability to promote permeability growth. The results also indicate that the sample size, L_a , has a crucial effect, in that it influences not only the scale of fluid flow but also the fracture behavior.

Despite the complex dynamics of volatile-consuming reactions in field settings, these results can inform the interpretation of the trends of permeability change observed in the presence of reactive processes such as hydration and carbonation. For example, Figure 4c provides a collection of the initial states of previously reported hydration and carbonation experiments. A hypothetical steady state threshold is also included purely for illustration purposes. According to our findings, the permeability is expected to decrease in serpentinization and peridotite carbonation experiments, which is consistent with the evidence reported by Farough et al. (2016) and Peuble et al. (2018). For natural serpentinization and peridotite carbonation, the permeability is instead expected to increase, which is also consistent with in situ investigations (Kelemen & Matter, 2008; O'Hanley, 1992). This correspondence between evidence and the synthetic results obtained here through a numerical model is encouraging. Nevertheless, it is important to remark that further in situ and laboratory measurements are still needed to confirm the possible existence of a steady state locus for site-specific conditions and accurately determine its location within the space of the control parameters defined in this work.

6. Conclusions

In this work, we first identified the dimensionless control parameters of volatile-consuming reactions by considering in conjunction the equations that govern reactive transport and rock fracture. Then, a discrete element model equipped with pore-scale multiphysics was used to investigate different regimes of hydro-chemo-mechanical behavior during this type of reaction by using the selected control parameters as the main markers of the systems' characteristics.

The numerical model effectively replicates permeability evolution trends observed in laboratory experiments by capturing the competition between pore clogging due to product formation and pore dilation due to fracturing. This implies that a critical value of porosity may exist where these competing processes balance each other, providing new insights into controlling permeability in the presence of volatile-consuming reactions such as hydration and carbonation. The spatial distributions of reaction extent, porosity, and normalized pore pressure across numerical samples were also examined to corroborate the finding that the local porosity eventually converges to a steady state.

Different regimes of permeability change are identified, and steady state loci are found in two non-dimensionalized control parameter spaces regulated collectively by porosity, mechanical properties (i.e., rock elasticity, fracture strength, and mineral expansion), hydro-chemical properties (rates of reaction and fluid flow), and system geometry. As a consequence, our results clearly show that steady state conditions are not only controlled exclusively by the reaction-transport dynamics, but that are also crucially constrained by rock volume change, elastic stress buildup, and, ultimately, fracture.

These findings extend and enrich the conclusions of previous laboratory studies, typically constrained by the inability to fully reach steady state conditions. In addition, they bear significant implications for subsurface carbon dioxide sequestration, in that they offer guidance to predict long-term permeability evolution and optimize the efficiency of carbon storage.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The output data of the numerical simulations are available at Dryad via Wu (2025).

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